

```

1  def first_fit(blocks, processes): 1 usage
2      print("\n-- First Fit ---")
3      allocation = [-1] * len(processes)
4
5      for i in range(len(processes)):
6          for j in range(len(blocks)):
7              if blocks[j] >= processes[i]:
8                  allocation[i] = j
9                  blocks[j] -= processes[i]
10                 break
11
12     print_result(allocation, processes)
13 def best_fit(blocks, processes): 1 usage
14     print("\n-- Best Fit ---")
15     allocation = [-1] * len(processes)
16
17     for i in range(len(processes)):
18         best = -1
19         for j in range(len(blocks)):
20             if blocks[j] >= processes[i]:
21                 if best == -1 or blocks[j] < blocks[best]:
22                     best = j
23
24         if best != -1:
25             allocation[i] = best
26             blocks[best] -= processes[i]
27
28     print_result(allocation, processes)
29 def worst_fit(blocks, processes): 1 usage
30     print("\n-- Worst Fit ---")

```

```

29 def worst_fit(blocks, processes): 1 usage
30     print("\n-- Worst Fit ---")
31     allocation = [-1] * len(processes)
32
33     for i in range(len(processes)):
34         worst = -1
35         for j in range(len(blocks)):
36             if blocks[j] >= processes[i]:
37                 if worst == -1 or blocks[j] > blocks[worst]:
38                     worst = j
39
40         if worst != -1:
41             allocation[i] = worst
42             blocks[worst] -= processes[i]
43
44     print_result(allocation, processes)
45 def print_result(allocation, processes): 3 usages
46     print("Process No.\tProcess Size\tBlock No.")
47     for i in range(len(processes)):
48         if allocation[i] != -1:
49             print(f"{i+1}\t\t{processes[i]}\t\t{allocation[i]+1}")
50         else:
51             print(f"{i+1}\t\t{processes[i]}\t\tNot Allocated")
52 blocks = [100, 500, 200, 300, 600]
53 processes = [212, 417, 112, 426]
54 first_fit(blocks.copy(), processes)
55 best_fit(blocks.copy(), processes)
56 worst_fit(blocks.copy(), processes)

```

--- First Fit ---

Process No.	Process Size	Block No.
1	212	2
2	417	5
3	112	2
4	426	Not Allocated

--- Best Fit ---

Process No.	Process Size	Block No.
1	212	4
2	417	2
3	112	3
4	426	5

--- Worst Fit ---

Process No.	Process Size	Block No.
1	212	5
2	417	2
3	112	5
4	426	Not Allocated

Process finished with exit code 0